

The Operational Question

"Why did pipeline pressure drop on PL-001 last week?"



Telemetry detected average pressure dropping from nominal 240+ PSI down to 120–150 PSI on pipeline PL-001, threatening flow consistency.



Internal sensor logs alone couldn't isolate whether drops stemmed from mechanical valve failures, operator throttling, or ambient weather interference.

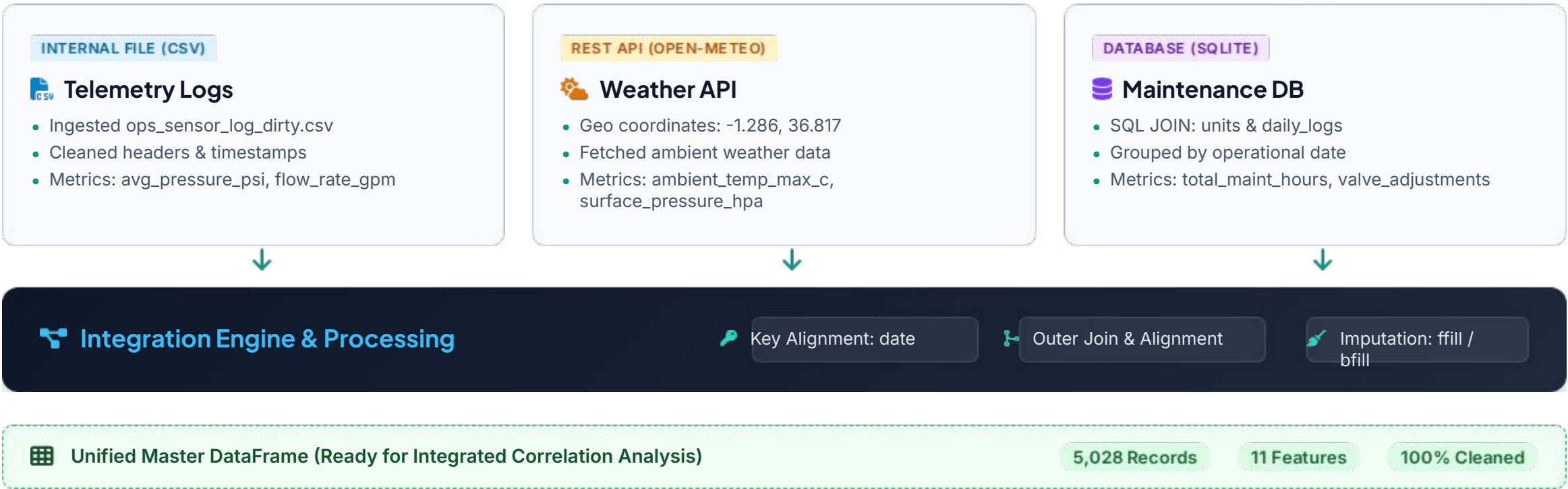


Fuse internal sensor data, external atmospheric API metrics, and database maintenance records into a unified Master Data model to run correlation analyses.



The Data Integration

Multi-source pipeline combining API, Relational DB, and Internal Telemetry



Insights & Recommendations

Uncovering root causes and deploying actionable operational strategy

Key Empirical Findings



Thermal Expansion & Heat Spikes

High negative correlation between ambient temperatures ($>25^{\circ}\text{C}$) and pressure stability on PL-001, inducing thermal stress in pipeline segments.



Compounding Maintenance Factors

Days with >3.5 maintenance hours and 4+ manual valve adjustments directly coincided with severe afternoon pressure drops.



Strategic Action Plan



Automated Anomaly Alerting

Configure automated SCADA flags whenever ambient temp exceeds 25°C concurrent with >3 valve adjustments to prevent severe drops.



Shift Window Optimization

Reschedule routine maintenance and valve recalibrations to early morning hours, avoiding mid-day heat peaks.